

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)**(TRIMESTER –II) END TERM EXAMINATION, JANUARY -2025****Quantitative Techniques****[Time: 2 Hours]****[Maximum Marks: 60]****Following are the course outcomes of the subject: - Quantitative Techniques**

CO	Course Outcome (CO)
CO1	Understand the fundamentals of quantitative techniques and its interdisciplinary role in decision making
CO2	Identify the right models for solving the particular problem
CO3	Apply the models for decision making and forecasting.

*(Note- Faculty can add the more CO's as per their subject.)***SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)**

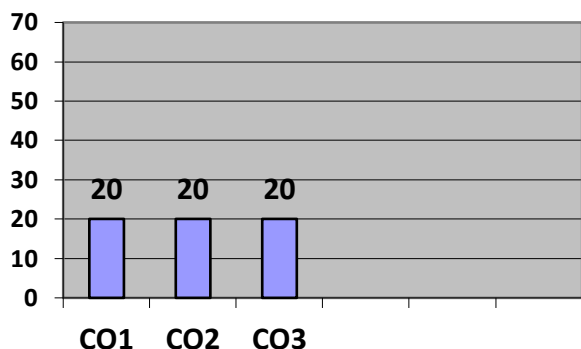
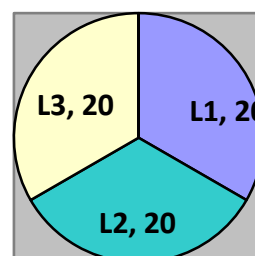
Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	4	CO - 1	BL - 1
(B)	4	CO - 1	BL- 1
(C)	4	CO - 1	BL-1
(D)	4	CO - 1	BL- 1
(E)	4	CO - 1	BL- 1

SECTION-B**Attempt ANY TWO questions out of the following: (Long Answer Type Questions, do not exceed 300- 400 words)**

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2 (A)	10	CO – 2	BL- 2
2 (B)	10	CO – 2	BL-2
3 (A)	10	CO – 3	BL-3
3 (B)	10	CO – 2	BL-2

SECTION-C**Attempt all parts of the following: (Caselet / Numerical, do not exceed 300- 400 words)**

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4 (A)	10	CO – 3	BL- 3
4 (B)	10	CO - 3	BL- 3

Course Outcome Wise Marks Distribution**Bloom's Level wise Marks Distribution****Bloom's Level wise Marks Distribution**

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GNIOT Institute of Management Studies – PGDM (Batch 2024-26)**(TRIMESTER –II) END TERM EXAMINATION, JANUARY - 2025****Quantitative Techniques****[Time: 2 Hours]****[Maximum Marks: 60]****INSTRUCTIONS:**

- (i) **There are THREE sections in this question paper.**
(ii) **Attempt questions from each section as per the directions.**
(iii) **Make suitable assumptions wherever necessary.**

SECTION-A**(Short Answer Type Questions, do not exceed 200 words)****Q.1. Attempt all parts of the following:****(5×4=20)**

- (a) If a researcher wants to understand the average income of employees at a large company with different departments, which type of sampling technique would be best, and why?
(b) A manager has to choose among three suppliers for raw materials, but there is no information about the probability of each supplier delivering on time. What non-probabilistic decision-making criterion could the manager use to make a decision under uncertainty?
(c) What is sampling error, and how can it be affected on the results of a survey?
(d) Explain the difference between a null hypothesis and an alternative hypothesis. Give an example of each in the context of testing the effectiveness of a new medicine on the diabetic patients.
(e) A researcher mistakenly rejects a null hypothesis when it is true. Which type of error has occurred, and what are the potential consequences for the study's conclusions?

SECTION-B**(Long Answer Type Questions)****Q.2. Attempt all questions of the following****(2X10=20)**

- (a) A deck of 52 cards is shuffled, and two cards are drawn one after the other with replacement. What is the probability that the first card is a King and the second card is a Queen?

OR

- (b) A coin is thrown 5 times. What is the probability of obtaining 4 or more Heads?

Q.3. (a) Fit a straight-line trend by the method of least square to the data given below:

Year	2008	2009	2010	2011	2012	2013	2014
Sales(m.tonnes)	9	11	13	12	14	15	17

- i) Plot these figures on a graph and show the trend line.
ii) Estimate the likely sales for the year 2016.

OR

- (b) Calculate the 5 yearly moving average for the following data of a number of commercial industrial failures in a country during 1999 to 2014.

Year	No. of failures	Year	No. of failures
1999	23	2007	9
2000	26	2008	13
2001	28	2009	11
2002	32	2010	14
2003	20	2011	12
2004	12	2012	9
2005	12	2013	3
2006	10	2014	1

SECTION-C

Q.4. Attempt all parts of the following:

(2×10=20)

- 4(a) A manufacturing firm is bringing out a new model in order to map out its advertising campaign to determine whether the model appeal depends on age group or not. The firm takes a random sample from person's attending a preview of the new model and obtained the results summarised below:

Age group	Under 25	25- 40	40 - 50	50 and over	Total
Liked the Car	146	78	48	28	300
Disliked the Car	54	52	32	62	200
Total	200	130	80	90	500

Test, whether the model appeals and age groups are independent.

- 4(b) The following data represent the number of units of production per day turned out by 5 different workers using 4 different types of machines:

Workers	Machine Type			
	A	B	C	D
1	44	36	48	38
2	48	40	50	44
3	37	38	40	36
4	45	34	45	32
5	40	44	50	40

Test:

- 1) Whether the mean productivity is the same for 4 different machine types.
- 2) Whether the 5 workers differ with respect to mean productivity.